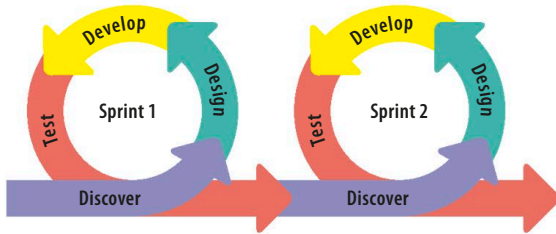


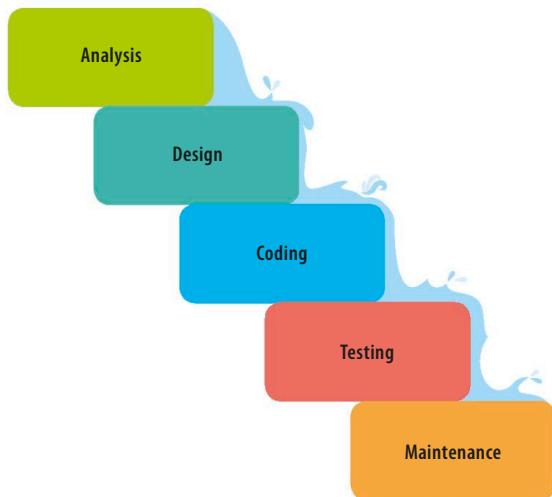
Managing groups

There are two main management styles for software projects. Agile development involves small, one- to two-week sections, called sprints. Every sprint contains the whole SDLC in miniature. Agile development allows teams to be flexible and responsive. Waterfall, by contrast, is a linear, traditional method. Each step is only done once, and in order.



△ Agile

Under the Agile style, there is an emphasis on working in small increments to set goals as opposed to planning. At the end of each sprint, a prototype is produced. Agile projects can react to changes in the client's requirements, or issues that pop up during development. The downside is that the development time is unpredictable. Plus, proper documentation is often neglected in favour of adding new features.



△ Waterfall

With the Waterfall style, it is easier to stay focused on the main goal. However, Waterfall style isn't flexible as the stages must happen in a set order. Also, the requirements of the client may change over time: a company may spend a few years developing code only to end up with software that doesn't suit the client's needs anymore.

Code collaboration

Multiple people can't edit the same file at the same time. Similarly, each file can't be restricted to a single developer, because many features overlap inside the code. But what happens if a change breaks part of the program? To resolve these issues, companies use multiple environments and special version-control software.

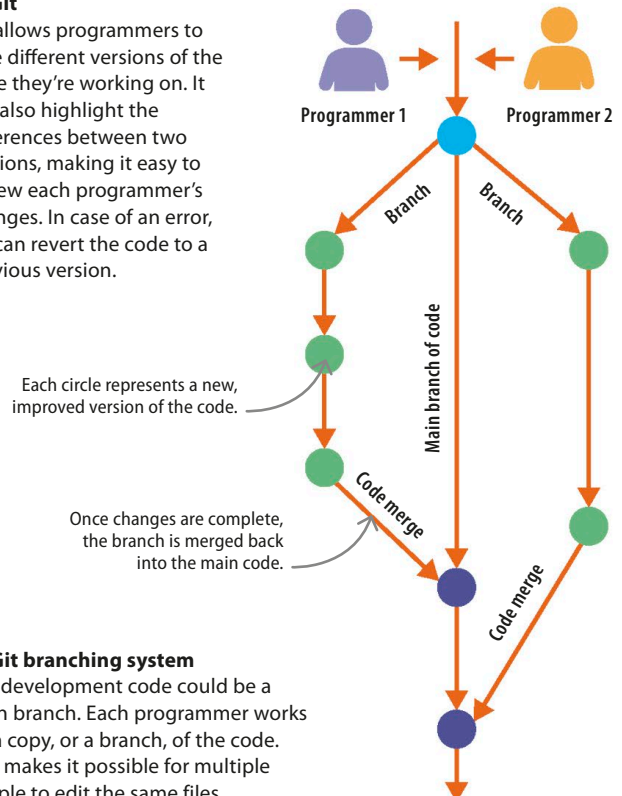
▽ Environments

Each programming environment has its own hardware and versions of its own code. Development contains code that programmers are currently editing. Staging contains a slightly older version whose features don't cause obvious crashes. Production contains the slick, professional, error-free code that's sold to customers.



▷ Git

Git allows programmers to save different versions of the code they're working on. It can also highlight the differences between two versions, making it easy to review each programmer's changes. In case of an error, Git can revert the code to a previous version.



▷ Git branching system

The development code could be a main branch. Each programmer works off a copy, or a branch, of the code. This makes it possible for multiple people to edit the same files.